

HEAT FIRST

A Geothermal Agricultural Services Enterprise

West Java, Indonesia

Investment Proposal

AC5699 – SUMAPS5150

1. Enterprise Overview

Heat First is a proposal for a heat-led geothermal enterprise located in the highland coffee and agricultural communities of West Java, Indonesia. The enterprise uses direct geothermal heat to provide farmers with coffee drying, cold storage and agro-processing services, and pairs these with a 1 MW binary cycle generator that supplies electricity to approximately 500 local households and sells excess power to PLN through a power purchase agreement. We are requesting a USD 2 million concessional loan from the Asian Development Bank (ADB) to co-finance this project alongside commercial debt and equity, and a long-term power purchase agreement (PPA) from PLN to provide the revenue certainty required to unlock that financing.

2. Introduction and Context

Indonesia holds an estimated 40% of global geothermal potential, driven by its location along the Pacific Ring of Fire.¹ West Java alone contains approximately 20% of that national capacity, with 62.2 MWth of direct-use heat resources distributed across established fields at Kamojang, Wayang Windu, Darajat and Patuha.² These fields are located in mountainous volcanic highlands where coffee, tea and vegetable farming are the primary economic activities, and where communities remain partially dependent on diesel for electricity and on sun-drying for agricultural processing. Despite this resources, geothermal development in Indonesia has largely taken the form of large-scale electricity generation, with few attempts to leverage geothermal heat directly for agricultural and community benefit.

Heat First is designed to close this gap. By targeting known surface geothermal manifestations and using shallow, low-enthalpy wells, the project reduces upfront capital requirements by approximately 80% compared to an explored alternative 5 MW electricity-first geothermal plant, while delivering more immediate and measurable impact to the farming communities it serves.

3. Problem Statement

Coffee farmers in the Pangalengan and Kamojang areas of West Java face two structural constraints that limit their incomes. First, they rely on sun-drying methods that are weather dependent, slow and produce inconsistent quality: full wash processing takes 7–10 days under conventional methods, and natural processing up to 35–42 days.³ Second, they have no access to cold storage infrastructure, which forces them to sell at harvest time when prices are lowest. Post-harvest losses in this context reach 30–40% of production.⁴ Both constraints reduce farmer incomes and limit their bargaining power in the value chain.

At the same time, approximately 500 households near these geothermal fields rely on diesel for electricity at an estimated cost of USD 0.25 per kWh, a significant energy burden. Reliable, affordable geothermal electricity would reduce household energy costs by approximately 76%, from roughly USD 325 to USD 78 per household per year.¹⁰

4. Products and Services

Heat First's primary products are heat-based agricultural services. A geothermal Dry House, modeled on the Geothermal Coffee Process (GCP) developed by Pertamina Geothermal Energy at Kamojang, uses geothermal steam to dry coffee, cutting full wash processing from 7–10 days to 2–4 days and natural processing from 35–42 days to 7–10 days.³ The resulting beans have stronger aroma, more fruity notes and smoother texture compared to conventionally processed beans, supporting premium pricing.⁶ The enterprise targets 100–150 tons of annual throughput in its first phase, with a service fee of approximately USD 0.50 per kg.¹⁰

A geothermal absorption chiller provides cold storage for local produce, reducing post-harvest losses from 30–40% to under 10% and allowing farmers to sell when market prices are favorable. Complementary agro-processing services, including tea drying, cocoa and spice drying, rice drying and mushroom cultivation sterilization, are already operating at nearby geothermal fields in West Java. Multi-criteria analysis by the Stanford Geothermal Workshop identifies coffee and rice drying as the most promising direct-use geothermal applications for Indonesia given temperature requirements, commodity demand and local resource availability.⁷

As a secondary output, a 1 MW binary cycle (ORC) generator produces electricity from the same geothermal resource. Approximately 91% of generation is sold to PLN at the applicable regulated tariff, with the remainder supplied to the local community at a community rate of USD 0.06 per kWh.¹⁰

5. Customers

The enterprise serves two customer groups. Primary customers are local farming households and cooperatives in the Pangalengan and Kamojang highlands who currently rely on sun-drying and have no access to cold storage. These farmers gain access to faster, higher-quality processing and storage infrastructure at a fee well below the income gains they can expect from reduced losses and premium pricing. West Java's coffee-growing highland districts produced approximately 6,499 tons of coffee in 2014, and production has grown since; the addressable market for heat services is therefore substantial relative to the enterprise's initial throughput targets.³

The secondary institutional customer is PLN, which purchases excess electricity through a power purchase agreement. While PLN revenue constitutes the larger share of total revenue in Year 1, the enterprise's social mission and primary value proposition are oriented toward the farming community, with heat services revenue growing as a share of total revenue over time as more farmers adopt the services.

6. Sources of Supply and Equipment

The heat services infrastructure (geothermal Dry House, absorption chiller, heat exchangers and piping) is directly adapted from Pertamina's existing GCP model at Kamojang, which provides a technical and procurement blueprint. The GCP has already demonstrated operational viability in the same region and with the same geothermal resource characteristics, substantially reducing equipment sourcing and commissioning risk.³

The electricity generation component uses a small modular binary cycle Organic Rankine Cycle (ORC) unit. This technology has been successfully deployed at small scale in Indonesia: at Lahendong in North Sulawesi¹⁵ and at Kamojang in West Java,¹⁶ confirming the availability of compatible equipment, local technical expertise and established service networks within Indonesia's geothermal sector.

Drilling services for the shallow wells required by this project are available from Indonesia's active geothermal drilling sector, concentrated in West Java where the Kamojang, Wayang Windu and Darajat fields have sustained long-term drilling activity.¹⁷ By targeting known surface manifestations with documented hot spring activity rather than undertaking speculative deep drilling, the enterprise significantly reduces both drilling cost and geological risk relative to a conventional geothermal project.²

7. Business Operations and Logistics

The enterprise operates two integrated systems from a single site near the geothermal resource. The heat facility runs on a service-based model: farming cooperatives deliver coffee and other agricultural produce to the Dry House facility, pay a processing fee, and collect finished product within the reduced drying timeframe. Cold storage operates on a per-day or per-ton rental basis, allowing farmers to store produce and time their sales. Both services are coordinated through existing cooperative structures in the Pangalengan and Kamojang areas, which reduces the logistical burden on individual smallholders.

The electricity generation component runs continuously at a capacity factor of approximately 85%, consistent with geothermal benchmarks.¹⁰ Grid connection serves the local community network, with excess generation fed to PLN's distribution system. Operations are managed by a small on-site team, with roles designed to be filled predominantly by local hires, consistent with community benefit-sharing practices in comparable geothermal projects and an important part of securing and maintaining social license.¹⁴

8. Business Strategy

Heat First's commercial strategy is built on two revenue streams that reinforce each other's viability. PLN electricity sales provide predictable, contracted baseload revenue that anchors the financial model and services debt, while heat services revenue grows organically as farmer adoption increases and the cooperative partnership deepens. This structure means the enterprise does not depend on either revenue stream alone: if heat services adoption is slower than projected, electricity revenue provides a stable floor; if heat services grow faster, returns improve above baseline.

On the sales side, the electricity revenue relationship is governed by the PPA with PLN, which is a condition precedent to financial close. Heat services are marketed directly through cooperative partnerships, with the

GCP model at Kamojang serving as a local proof of concept that reduces the need for extensive farmer education or market development.³ Carbon credits provide an additional revenue layer, with the enterprise targeting international voluntary carbon markets, where comparable renewable energy projects have attracted USD 15–20 per ton, against a conservative baseline of USD 10 per ton used in the financial model.

Community relations and local employment are not treated as an afterthought but as core to the enterprise's operating model. Local hiring, transparent fee structures and benefit-sharing with cooperatives are built into the project design from the outset. Experiences from geothermal projects in comparable emerging market contexts show that community acceptance is a material determinant of long-term project stability, and that embedding social benefit structures early substantially reduces this risk.¹⁴

9. Regulatory Environment

Indonesia's electricity sector is organized around PLN as the central buyer and distributor. Private developers generate electricity and sell it to PLN through power purchase agreements under Ministry of Energy and Mineral Resources regulation.⁸ For plants under 10 MW, Presidential Regulation No. 112/2022 sets a ceiling tariff of 9.76 cents per kWh multiplied by a regional location factor; on Java this factor is 1.0, giving an effective tariff of USD 0.097 per kWh, the rate used in the financial model.⁹ MEMR Regulation No. 5/2025 sets a maximum PPA duration of 30 years for geothermal projects, consistent with the project horizon used in the financial model. Draft tariff revisions currently propose increasing the rate for 1–10 MW plants to USD 0.145 per kWh, which would materially improve project returns if confirmed.

For direct use of geothermal heat, the regulatory framework is less codified but increasingly supportive, with recent policy emphasis on agricultural and industrial direct-use applications. On carbon credits, the financial model uses a conservative baseline of USD 10 per ton of CO₂ displaced, consistent with domestic Indonesian carbon exchange pricing, with upside potential if the enterprise accesses international voluntary carbon markets, where comparable renewable energy and avoided-emissions projects have historically attracted USD 15–20 per ton from buyers under frameworks such as Verra or Gold Standard. Whether Indonesian geothermal projects can access these international markets at scale depends on evolving carbon trading policy, which remains in development.

10. Competitive Landscape

The competitive landscape for this enterprise is thin at the scale and service model it proposes. Most geothermal development in Indonesia, including Pertamina's core operations and PLN's generation portfolio, operates at utility scale and focuses on grid-scale electricity generation, without offering bundled heat services to agricultural communities. That said, Pertamina itself has pioneered the closest existing model to what Heat First proposes: the Geothermal Coffee Process (GCP) at Kamojang, operated in partnership with local entrepreneur Mang Deden, which currently partners with over 80 farmers, processed 20 tons of coffee last season, and has shown both quality improvements and significantly faster processing times.³ The GCP is therefore best understood not as a competitor but as a proof of concept and potential strategic partner, thereby validating the heat services model from within the same sector. Heat First builds on this precedent by adding cold storage, broadening the agro-processing service range and integrating a small electricity component, elements the GCP does not currently offer.

The alternative for heat services would be sun-drying, which is free but weather-dependent, slow and produces inconsistent quality. Cold storage has no local competitor at all, its absence is precisely the market gap the enterprise addresses. For electricity, the alternative is diesel generation at USD 0.25 per kWh, against which the community tariff of USD 0.06 per kWh is strongly competitive. There is no other enterprise currently offering this bundled heat + cold storage + electricity model at small scale in this region.

11. Financial Plan

Total project capital expenditure is USD 4.75 million, approximately 80% lower than a comparable 5 MW electricity-first geothermal plant.¹⁰ The capital structure combines a USD 2 million ADB concessional loan with USD 975,000 in commercial debt and USD 1.275 million in equity, against a net CapEx of USD 4.25 million after a USD 500,000 exploration grant sought separately from Indonesian government geothermal support programs. The blended interest rate across the debt stack is approximately 5.66%, with annual debt service of USD 299,456 over a 15-year repayment term.¹¹

Year 1 total revenue is projected at approximately USD 918,000, with electricity sales to PLN representing the largest share, followed by heat services, coffee drying, cold storage and agro-processing, and carbon credits. The enterprise is profitable from Year 1, with EBITDA of approximately USD 518,000 and a cash-

on-cash return on equity of approximately 17%. The Debt Service Coverage Ratio in Year 1 is 1.73x, rising each year as revenues grow. Equity breakeven occurs in Year 6, and upon full loan repayment in Year 15, net income nearly doubles. Full 30-year projections are provided in Attachment 2.¹²

12. Financial, Social and Environmental Impacts

The social impact of Heat First is embedded in the heat-first model itself. Coffee farmers gain access to a processing facility that produces higher-quality beans in a fraction of the time, reducing weather-related income volatility and unlocking premium market prices. Cold storage reduces post-harvest losses from 30–40% to under 10%,⁴ allowing farmers to time sales strategically and improve their bargaining position. For electricity, 500 households spending approximately USD 325 per year on diesel would pay approximately USD 78 per year at the community tariff, a 76% reduction in household energy costs.¹⁰ Geothermal development more broadly is associated with local employment generation and more stable long-term energy pricing, both of which disproportionately benefit lower-income rural households.¹³

Experiences from comparable geothermal projects in Kenya demonstrate that benefit-sharing mechanisms and local employment commitments are material determinants of community acceptance and long-term project stability.¹⁴ Heat First builds these structures through cooperative partnerships, local hiring and transparent fee structures rather than treating them as add-ons.

Environmentally, the project displaces diesel-based electricity generation and fossil fuel-dependent agricultural processing, generating annual carbon credit revenue modeled conservatively in Attachment 2 and with meaningful upside potential as discussed in Section 9.¹² The project also contributes to Indonesia's renewable energy transition targets by adding reliable baseload geothermal generation to the West Java grid without requiring PLN capital investment.

13. Risks and Mitigation

Geological risk. By targeting known surface manifestations rather than undertaking speculative deep drilling, the enterprise significantly reduces subsurface uncertainty. West Java has over 130 documented surface manifestations, and the exploration budget focuses on well confirmation rather than greenfield exploration.²

Regulatory and off-take risk. The primary regulatory risk is uncertainty around future PLN tariff revisions. The financial model uses the current conservative tariff of USD 0.097 per kWh; any upward revision improves returns.⁹ A signed PPA with PLN is a condition precedent to financial close.

Market adoption risk. Heat services revenue depends on farmer adoption. This risk is substantially mitigated by the existing GCP model at Kamojang, which already demonstrates commercial viability with 80+ farmer partners in the same region,³ and by formal cooperative partnerships to be established as anchor customers before financial close.

Financing risk. High upfront CapEx relative to available concessional capital is the primary financing constraint for small geothermal projects globally.¹⁸ The blended capital structure is designed to reduce the cost of capital to a level compatible with the project's revenue profile, and the DSCR of 1.73x in Year 1 provides meaningful downside buffer for lenders.¹²

Reputational and social risk. Local community opposition can delay or derail geothermal projects, particularly where drilling raises concerns about seismicity or groundwater. The heat-first model, which starts with services that directly benefit the community, creates a fundamentally different relationship with local stakeholders compared to a conventional electricity plant, reducing this risk materially.¹⁴

14. Key Steps and Implementation Timeline

The enterprise builds on a strong precedent base that substantially de-risks the development pathway. The GCP at Kamojang has demonstrated geothermal heat-based coffee drying in West Java,³ the Wayang Windu field has confirmed coffee drying using geothermal brine at 80°C through a closed-loop heat exchanger,⁵ and small-scale binary ORC generation is operating at both Lahendong¹⁵ and Kamojang.¹⁶ The enterprise is not starting from scratch: it is scaling what has already been shown to work.

Phase 0, Development and Financial Close (Months 1–12). Immediate priorities: negotiating and signing a PPA framework agreement with PLN; completing surface confirmation drilling at the target site in the Pangalengan or Kamojang area; finalizing the ADB concessional loan facility; and establishing formal offtake partnerships with local coffee cooperatives. Environmental and social impact assessment and permitting proceed in parallel.

Phase 1, Construction and Commissioning (Months 12–24). Civil works, well completion, ORC unit installation and heat infrastructure commissioning. First coffee drying services to cooperative partners begin toward the end of this phase, ahead of full commercial operations.

Phase 2, Operations and Expansion (Year 2 onwards). Full commercial operations start. Heat services grow as additional cooperatives join. Carbon credit registration and first credit sales. The enterprise monitors geothermal resource performance and evaluates the feasibility of adding a second ORC module if resource and demand conditions support it. The model is designed to be replicable across other highland geothermal areas in West Java and beyond.¹⁹

15. Request for Resources

From the Asian Development Bank: a USD 2 million concessional loan at a subsidized interest rate, disbursed at financial close, with a 15-year repayment term. This concessional tranche is essential to bring the blended cost of capital below the project's hurdle rate and to crowd in commercial debt and equity. ADB's participation would also provide institutional credibility that supports engagement with PLN and local partners. The project's social and environmental profile, rural income support, reduced diesel dependence, post-harvest loss reduction and geothermal renewable energy, aligns directly with ADB's mandate in Indonesia's energy and agriculture sectors.

From PLN: a 20-year power purchase agreement for electricity generated by the 1 MW binary plant, at the applicable tariff under Presidential Regulation No. 112/2022, well within the 30-year regulatory ceiling under MEMR Regulation No. 5/2025. A defined term of at least 20 years is standard practice in geothermal project finance, as it provides lenders with revenue certainty over the full loan repayment period and into the project's operating life. PLN's off-take commitment on these terms is a condition precedent to financial close. In return, PLN gains reliable baseload geothermal supply from a small-scale developer operating within the regulatory framework, contributing to renewable energy targets without requiring PLN capital investment.

Together, these commitments would enable financial close on a project that is commercially viable from Year 1, generates strong and growing returns over a 30-year operating life, and delivers measurable social and environmental impact in one of Indonesia's most geothermal-rich agricultural regions.

References

- ¹ IEA (2023). The Future of Geothermal Energy. Paris: IEA. [Indonesia holds ~40% of global geothermal potential.]
- ² GeoCap Program (2019). Direct use of geothermal resources in Indonesia. <https://www.geocap.nl/handbook/direct-use/> [Documents 62.2 MWth of direct-use heat potential across West Java and Banten, and over 130 surface manifestations.]
- ³ ThinkGeoEnergy (2025). Pioneering geothermal use for coffee production in Kamojang, Indonesia. <https://www.thinkgeoenergy.com/pioneering-geothermal-use-for-coffee-production-in-kamojang-indonesia/> [GCP: 80+ farmer partners, 20 tons processed per season; full wash drying reduced from 7–10 days to 2–4 days; natural processing from 35–42 days to 7–10 days.]
- ⁴ FAO (2021). Transform food systems to avert \$400 billion annually in loss and waste. [Post-harvest losses in developing countries reach 30–40% of production without cold chain infrastructure.]
- ⁵ Prasetyo, R.M., Wicaksono, A., Biddinika, M.K. & Takahashi, F. (2018). Study of geothermal direct use for coffee drying at Wayang Windu geothermal field. AIP Conference Proceedings 2026(1):020029. DOI: 10.1063/1.5064989. [Geothermal brine at 80°C dries coffee from 60% to 12% water content in 12.5 hours using a closed-loop heat exchanger at Wayang Windu, West Java.]
- ⁶ ThinkGeoEnergy (2023). Geothermal heat being used for coffee processing in Indonesia. <https://www.thinkgeoenergy.com/geothermal-heat-being-used-for-coffee-processing-in-indonesia/> [GCP beans: stronger aroma, more fruity notes, smoother texture compared to conventionally sun-dried beans.]
- ⁷ Bagaskara et al. (2023). Exploring New Ideas to Improve Geothermal Direct Use in Indonesia. Stanford Geothermal Workshop. <https://pangea.stanford.edu/ERE/db/GeoConf/papers/SGW/2023/Bagaskara2.pdf> [Multi-criteria decision analysis identifies coffee and rice drying as the two most promising direct-use geothermal applications in Indonesia, based on temperature requirements, commodity demand and local resource availability.]
- ⁸ World Bank (2017). Indonesia's Power Sector: Current Status and Challenges. <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/762021486868465688> [PLN is the exclusive buyer and distributor of electricity; private developers sell via PPAs under MEMR regulation.]
- ⁹ International Comparative Legal Guides (ICLG). (2024). Renewable Energy Laws and Regulations – Indonesia. <https://iclg.com/practice-areas/renewable-energy-laws-and-regulations/indonesia> [PR 112/2022: ceiling tariff 9.76 cents/kWh × location factor; Java factor = 1.0, giving USD 0.097/kWh for plants under 10 MW. MEMR Reg. 5/2025: 30-year maximum PPA term for geothermal projects.]
- ¹⁰ See Attachment 2 – Quantitative Financial Model (Indonesia_1MW_HeatFirst_Geothermal.xlsx), Assumptions Sheet. [Key assumptions: capacity factor 85%; total CapEx USD 4.75M (~80% below 5 MW electricity-first model at USD 24.5M); coffee drying service fee USD 0.50/kg; community household savings from USD 325/yr at diesel to USD 78/yr at USD 0.06/kWh community tariff.]
- ¹¹ See Attachment 2 – Quantitative Financial Model, Financing Plan Sheet. [Capital structure: USD 2M ADB concessional loan + USD 975k commercial debt + USD 1.275M equity; net CapEx after USD 500k exploration grant: USD 4.25M; blended interest rate: 5.66%; annual debt service: USD 299,456 over 15 years.]
- ¹² See Attachment 2 – Quantitative Financial Model, Pro Forma P&L and Break-Even Sheets. [Year 1 total revenue: ~USD 918k; Year 1 EBITDA: ~USD 518k; cash-on-cash return on equity: ~17%; DSCR Year 1: 1.73x; equity breakeven: Year 6; loan fully repaid: Year 15, after which net income nearly doubles.]
- ¹³ World Bank (2023). Geothermal Energy: Unveiling the Socioeconomic Benefits. Washington, DC: World Bank. <https://documents1.worldbank.org/curated/en/099122823090547278/pdf/P1744881ab11080191a03411d191385e065.pdf> [Geothermal development generates local employment and more stable long-term energy pricing, with distributional benefits for lower-income households.]
- ¹⁴ Abdi, A.M., Murayama, T., Nishikizawa, S., Suwanteep, K. and Mariita, N.O. (2024). 'Determinants of community acceptance of geothermal energy projects: A case study on a geothermal energy project in Kenya.' Renewable Energy Focus, 50, 100594. [Community acceptance is materially shaped by benefit-sharing mechanisms and local employment commitments embedded early in project design.]

- ¹⁵ Richter, A. (2021). Small-scale binary geothermal plant ready to operate at Lahendong. <https://www.thinkgeoenergy.com/small-scale-binary-geothermal-plant-ready-to-operate-at-lahendong/> [Confirms commercial deployment of a small-scale binary ORC unit at Lahendong, North Sulawesi, Indonesia.]
- ¹⁶ Prasetyo, B.T., Suyanto, Cahyadi, & Sutriyanto, H. (2021). Lesson learned – The operation of the pilot scale geothermal power plant 3MW – Kamojang, Indonesia. *Geothermics*, 91. <https://www.sciencedirect.com/science/article/abs/pii/S0375650520303175> [Kamojang pilot plant operated at 3 MW; confirms small-scale geothermal electricity generation is viable in West Java.]
- ¹⁷ AECOM (2016). Wayang Windu Geothermal Power Plant, West Java. <https://aecom.com/projects/wayang-windu-geothermal-power-plant/> [Wayang Windu is one of several active geothermal fields in West Java sustaining long-term drilling activity and established local supply chains.]
- ¹⁸ World Bank (2021). Preparing Feasibility Studies for the Financing of Geothermal Projects: An Overview of Best Practices. Washington, DC: World Bank. <https://documents1.worldbank.org/curated/en/733791626931398765/pdf/Preparing-Feasibility-Studies-for-the-Financing-of-Geothermal-Projects-An-Overview-of-Best-Practices.pdf> [High upfront CapEx and geological risk are the primary financing constraints for geothermal projects; concessional finance and risk-sharing mechanisms are essential to project viability.]
- ¹⁹ McBride, J., Helmecci, J., Goh, A. and Mangalmurti, D. (2025). Unlocking Global Geothermal Energy: Pathways to Scaling International Deployment of Next-Generation Geothermal. Washington, DC: Carnegie Endowment for International Peace. [Discusses replicability of small-scale and community-integrated geothermal models across emerging markets including Indonesia.]